SAT Math · Test III

1 ★ Mark for Review ABC

A packaging engineer is optimizing a cardboard sleeve whose width profile is modeled by the quadratic expression $ax^2 + 176x + c$, where a and c are positive constants. If $px + q$ is a factor of the expression,

$$ax^2 + 176x + c = (px + q)(rx + s),$$

where p and q are positive constants, what is the greatest possible value of ac?

2 ★ Mark for Review **ABC**

A signal-processing filter uses a polynomial in the variable z^9 to tune resonance. In the given expression,

$$34z^{18} + bz^9 + 30,$$

b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b?

3 ★ Mark for Review ABC

An optics lab rewrites a focusing model into paired quadratic factors in x^2. The expression

$$6x^4 + 17x^2 + 7$$

can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive non-integers.

What is the value of $a + c$?

4 ★ Mark for Review **ABC**

A materials scientist factors a stiffness polynomial before running a simulation. The given quadratic function

$$54x^4 + 219x^2 + 105$$

has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d, k are integers, what is the smallest possible value of ab?

5 ★ Mark for Review ABC

A drone's altitude path over a short flight is modeled by a parabola in the xy-plane. The equation of a parabola is written in the form

$$y = ax^2 + bx + c,$$

where a, b, c are constants. When graphed in the xy-plane, the parabola has vertex $(-1, 4)$ and does not intersect the x-axis. Which of the following could be the value of $a - b - c$?

A. 2

B. 4

C. -1

D. -6

6 ★ Mark for Review **ABC**

A fountain's water arc is modeled by a downward-opening quadratic with vertex $(1, 32)$. A quadratic function with vertex $(1, 32)$ can be written as

$$-ax^2 + bx + c \quad (a, b, c \text{ are all positive integers}).$$

What is the maximum value of b?

7 ★ Mark for Review ABC

A robotics team measures that a test cart's speed is increasing at a rate of 7.7 meters per second squared. What is this rate in **miles per minute squared**, rounded to the nearest tenth? (Use 1 mile = 1609 meters.)

A. 0.3

B. 17.2

C. 206.5

D. 209

8 ★ Mark for Review **ABC**

A bacteria culture in a bioreactor is modeled by an exponential growth function. The function f is defined by

$$f(x) = ab^{x/n},$$

where a, b, n are constants, and b, n are integers. If $f(2) = 6$ and $f(4) = 150$, what is the value of $f(5)$?

9 ★ Mark for Review **ABC**

A roller-coaster design uses a parabolic track segment $y = f(x)$. The graph of $y = f(x)$ in the xy-plane has a vertex at $(4, -7)$, contains the point $(3, -9)$, and has a y-intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y-intercept at $(0, b)$. What is the positive difference between a and b?

10 ★ Mark for Review ABC

An economist fits a quadratic revenue model $y = f(x)$ that breaks even at two production levels. The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a, b, c are constants. The graph of $y = f(x)$ passes through the points $(9, 0)$ and $(-2, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

A. 8

B. 7

C. -6

D. -12

11 ★ Mark for Review ABC

A bridge support cable clearance is checked using polynomial factors in x. Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 22x + 14b$

C. $3x^2 + 30x + 14b$

D. $3x^2 + 37x + 14b$

12 ★ Mark for Review ABC

A survey crew measures a triangular lot XYZ for a small park. In triangle XYZ, the measure of angle Y is 38°. The length of XY is 12 meters, and the length of YZ is 5 meters. What is the area, in square meters, of triangle XYZ?

A. 30

B. 60

C. $30 \sin 38^\circ$

D. $60 \sin 38^\circ$

13 ★ Mark for Review ABC

A quality team selects two random samples to estimate the average lifespan of a streetlamp bulb (in hours). Based on the first sample, the estimated average lifespan is 800 hours, with a margin of error of 20 hours.

Based on the second sample, the estimated average lifespan is 810 hours, with a margin of error of 35 hours.

Assuming the margins of error were calculated in the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

A. The first sample contained fewer light bulbs than the second sample.

B. The first sample contained more light bulbs than the second sample.

C. The first sample had a shorter average lifespan than the second sample.

D. The first sample had a longer average lifespan than the second sample.

14 ★ Mark for Review ABC

A vibration model for a machine mount is given by a quartic that is written in shifted form. The function $p(x)$ is defined by:

$$p(x) = a((x-3)^2 + b)((x-3)^2 - c),$$

where $a, b,$ and c are constants. The graph of $y = p(x)$ passes through the points $(1, 25)$ and $(-1, 52)$.

What is the value of $p(5) + p(7)$?

A. 25

B. 27

C. 52

D. 77

15 ★ Mark for Review ABC

A savings balance is modeled by an exponential function of time. In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$.

Which expression gives the value of c in terms of b?

$$f(x) = 66(b)^x$$

A. $1 + \frac{b}{100}$

B. $b + 100$

C. $100(b + 1)$

D. $100(b - 1)$

16 ★ Mark for Review ABC

A park service tracks the average growth of a young oak. On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

A. $\frac{29m}{4k}$

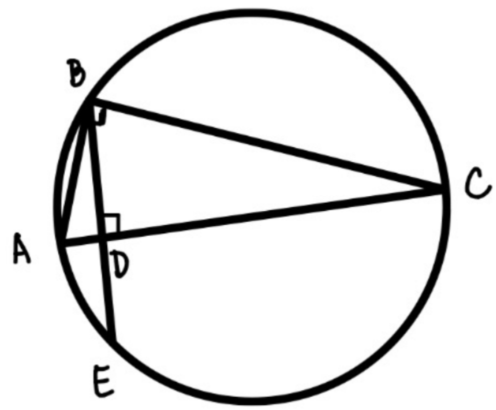
B. $\frac{29k}{4m}$

C. $\frac{1044m}{k}$

D. $\frac{1044k}{m}$

17 ★ Mark for Review **ABC**

A circular plaza is being surveyed. In the figure shown, points A, B, C, and E lie on the circle, and $AB < BC$. Angle ABC is a right angle, segment AC is a diameter of the circle, and segment AC is perpendicular to segment BE at point D. Also, $BD = \sqrt{346}$. The diameter of the circle is 175. If $\dfrac{CD}{AD} = r$, what is the value of r?



18 ★ Mark for Review ABC

An engineer solves a timing equation before calibrating two sensors. In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a?

$$(x - k)^2 = (k - 5a)(x - k)$$

A. $-\frac{32}{5}$

B. $\frac{32}{5}$

C. $-\frac{5}{32}$

D. $\frac{5}{32}$

19 ★ Mark for Review ABC

A lab predicts bacteria population (in thousands) with an exponential model. The equation $N(m) = 63(Q)^{m/6}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds.

What is the value of p in terms of Q?

A. $100(Q^{30} - 1)$

B. $100(Q^{\frac{1}{2}} - 1)$

C. $100(Q^{30} + 1)$

D. $100(Q^{1.5} + 1)$

20 ★ Mark for Review ABC

A logistics planner solves an equation whose solutions represent two delivery windows. In the given equation, m and n are constants. The sum and product of the solutions of the equation are 8 and $\frac{7}{2}$, respectively.

What is the value of $\frac{m}{n}$?

$$(x-m)^2 = (m+2n)(2x-n)$$

A. $-\frac{1}{3}$

B. $\frac{1}{3}$

C. -3

D. 3

21 ★ Mark for Review ABC

A landscaping crew is staining a yard fence and must buy enough product for two coats. One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S, in gallons, needed to stain the fence in this yard twice?

A. $S = \dfrac{w}{170}$

B. $S = 170w$

C. $S = 340w$

D. $S = \dfrac{w}{85}$

22 ★ Mark for Review ABC

A projectile's height in a scaled coordinate model is given by $y = -x^2 + 9x - 100$. In the xy-plane, the graph of this equation intersects the line $y = c$ at exactly one point. What is the value of c?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$